

Jonathan S. Sterling

Principal .NET Backend Architect

+1-555-010-9821 | j.sterling.dev@email.com

LinkedIn: linkedin.com/in/jsterling-dotnet | Location: Charlotte, NC

PROFESSIONAL SUMMARY

Visionary Senior .NET Backend Developer and Architect with over 9 years of hands-on experience in high-concurrency distributed systems, cloud-native application design, and enterprise-grade microservices. Expert in the .NET ecosystem (.NET 8/6, Core 3.1) with a deep mastery of C# performance optimization, memory management, and asynchronous programming patterns.

Throughout my career, I have successfully led the transformation of legacy monolithic architectures into scalable, event-driven microservices for Fortune 500 companies in the FinTech and Healthcare sectors. I specialize in Domain-Driven Design (DDD), Clean Architecture, and CQRS, ensuring that business logic remains decoupled from infrastructure. My technical leadership has resulted in measurable improvements in system throughput (40%+), significant reductions in cloud operational expenditure (30%+), and the establishment of high-velocity DevOps cultures. I am committed to writing clean, maintainable, and testable code while mentoring the next generation of software engineers.

CORE COMPETENCIES

- **Architectural Patterns:** Microservices, Event-Driven Architecture, Serverless, CQRS, Hexagonal/Onion Architecture.
- **Backend Development:** Advanced C# (LINQ, Generics, Reflection, Async/Await), Web API, gRPC, SignalR.
- **Data Management:** Relational (SQL Server, PostgreSQL), NoSQL (CosmosDB, MongoDB), Advanced Indexing.
- **Distributed Messaging:** Azure Service Bus, RabbitMQ, Kafka, Event Grid.
- **Cloud & DevOps:** Azure Cloud (PaaS, IaaS), Docker, Kubernetes, Terraform, Azure DevOps CI/CD.
- **Testing & Security:** TDD/BDD, xUnit, Moq, OAuth2/OpenID Connect, HIPAA/SOC2 Compliance.

EXHAUSTIVE TECHNICAL SKILL MATRIX

3.1 Development Ecosystem

Frameworks	.NET 8, .NET 6, .NET Core 3.1/2.1, .NET Framework 4.8/4.7.2, ASP.NET Core MVC.
ORM & Data Access	Entity Framework Core (7/8), Dapper, ADO.NET, AutoMapper, FluentValidation.
Communication	RESTful APIs (HATEOAS), gRPC (Protobuf), WebSockets, SignalR Core.
Languages	C# (Expert), SQL (T-SQL, PL/pgSQL), PowerShell, TypeScript (Basic), Bash.

3.2 Cloud Infrastructure (Microsoft Azure)

Compute	App Services, Azure Functions (V3/V4), Azure Container Apps, AKS (Kubernetes).
Storage & Data	Azure SQL Database, CosmosDB (SQL/Gremlin), Blob Storage, Table Storage.
Messaging	Service Bus (Topics/Subscriptions), Queue Storage, Event Hubs, Event Grid.
Security & Config	Azure Key Vault, Managed Identities, App Configuration, Entra ID (Azure AD).

3.3 Persistence & Performance

RDBMS	Microsoft SQL Server (2016-2022), PostgreSQL (12-16), MySQL.
NoSQL	MongoDB, Redis (Distributed Caching), Elasticsearch (ELK Stack).
Performance	SQL Profiler, Execution Plan Analysis, BenchmarkDotNet, dot-Memory, dotTrace.

3.4 DevOps, Tooling & Methodology

CI/CD	Azure Pipelines (YAML), GitHub Actions, Jenkins, Octopus Deploy.
Containers	Docker (Multi-stage builds), Docker Compose, Helm Charts.
Infrastructure	Terraform, Bicep, ARM Templates.
Observability	Application Insights, Prometheus, Grafana, Serilog, New Relic.
Agile Tools	Jira, Azure Boards, Confluence, Trello.

PROFESSIONAL EXPERIENCE

FinTech Global Solutions

Lead Backend Architect / Senior Engineer

Jan 2021 – Present
Remote / Charlotte, NC

Overview: Leading the core engineering group responsible for the high-frequency transaction engine servicing over 12 million global users.

Key Achievements & Responsibilities:

- **Microservices Transformation (2021-2022):** Led the 18-month strategic migration of a legacy .NET 4.7 monolith to a .NET 8 microservices ecosystem. Utilized Strangler Fig pattern to decouple services without downtime.
- **Scalability:** Designed an asynchronous ledger update system using **Azure Service Bus** that reduced peak-load database contention by 70% and increased throughput from 1,200 to 8,000 transactions per second.
- **Cost Optimization:** Conducted a comprehensive audit of Azure SQL Elastic Pools and transitioned low-access data to CosmosDB with TTL (Time-to-Live) policies, saving the firm \$14,000 per month in storage costs.
- **Performance Tuning:** Identified a critical bottleneck in the settlement service; by refactoring Entity Framework queries into Dapper and optimizing SQL execution plans, reduced P99 latency from 1.2s to 180ms.
- **Security Governance:** Implemented a Zero Trust architecture using Managed Identities and Key Vault, ensuring that no production connection strings were stored in code or environment variables.
- **Leadership:** Directly manage and mentor a cross-functional team of 10 (8 Developers, 2 QA). Established a peer-review protocol that increased unit test coverage from 45% to 92% across all new services.

HealthDirect Systems

Senior .NET Software Developer

June 2017 – Dec 2020
Nashville, TN

Overview: Specialized in building HIPAA-compliant backend services for a major healthcare provider, focusing on real-time data ingestion.

Key Achievements & Responsibilities:

- **Real-time Ingestion:** Engineered a serverless data ingestion pipeline using **Azure Functions** and **Event Hubs** to process telemetry from 250,000+ IoT wearable devices.
- **Caching Strategy:** Architected a multi-level caching system using Redis and in-memory LazyCache to handle patient lookup surges during COVID-19 vaccine rollouts, preventing system crashes during 500% traffic spikes.
- **Clean Architecture:** Introduced "Clean Architecture" templates (Core, Application, Infrastructure, Web) to the engineering department, which standardized service development and reduced "ramp-up" time for new hires by 40%.
- **Compliance:** Automated the encryption-at-rest and encryption-in-transit protocols for PII/PHI data, ensuring 100% compliance with HIPAA audits in 2018 and 2019.

Overview: Mid-level developer role focusing on enterprise retail management software for mid-to-large scale distributors.

Key Achievements & Responsibilities:

- **Reporting Engine:** Developed a custom data aggregation engine in C# that synthesized sales data from 12 disparate sources into a unified SQL Data Warehouse. Reduced month-end reporting time from 3 days to 4 hours.
- **UI Integration:** Built responsive admin dashboards using ASP.NET MVC and Bootstrap, allowing non-technical managers to monitor inventory levels in real-time.
- **API Modernization:** Assisted in the transition from SOAP/WCF services to RESTful Web APIs, enabling third-party vendor integrations that grew the company's partner ecosystem by 25%.
- **Database Optimization:** Handled the normalization of a legacy 400-table database schema, improving general application responsiveness by 30%.

DETAILED PROJECT PORTFOLIO

5.1 Case Study 1: Global Multi-Tenant E-Commerce Core

Objective: Build a "Headless" e-commerce backend capable of supporting 50+ localized brands with unique tax, currency, and inventory requirements.

Technical Deep-Dive:

- **Technology Stack:** .NET 6, ASP.NET Core, PostgreSQL, Redis, Docker, AKS.
- **Solution:** Implemented a "Database-per-Tenant" strategy for high-value clients and custom middleware in C# to resolve the tenant ID from host headers, injecting correct DB context at runtime.
- **Impact:** Enabled the launch of 5 new international brands in 3 weeks. Handled \$450M in annual GMV with zero multi-tenant data leaks.

5.2 Case Study 2: Real-time Logistics Fleet Monitor

Objective: Provide real-time location and diagnostics tracking for a fleet of 15,000 heavy-duty vehicles.

Technical Deep-Dive:

- **Technology Stack:** .NET Core 3.1, SignalR, Azure IoT Hub, CosmosDB.
- **Solution:** Utilized a "Lambda Architecture" for hot data visualization via SignalR and parallel archiving to CosmosDB for cold storage analysis.
- **Impact:** Reduced "Lost Vehicle" incidents by 22% and saved the client \$1.2M in annual fuel costs via efficiency insights.

5.3 Case Study 3: Centralized Identity & Access Platform

Objective: Replace multiple disparate login systems with a single, secure Identity Provider (IdP) for 20,000 employees.

Technical Deep-Dive:

- **Technology Stack:** .NET 6, IdentityServer4, EF Core, Azure Key Vault.
- **Challenge:** Consolidating user data from AD and SQL while ensuring SSO across 30 apps.
- **Impact:** Reduced identity-related helpdesk tickets by 55% and enforced MFA across the entire enterprise ecosystem.

5.4 Case Study 4: Distributed Batch Processing Service

Objective: Migrate legacy Windows Service-based nightly batch jobs to a cloud-native, scalable architecture.

Technical Deep-Dive:

- **Technology Stack:** .NET 8, Azure Functions, Azure Queue Storage.
- **Solution:** Re-engineered the process into a Producer-Consumer pattern with horizontal scaling, processing 2M records nightly.
- **Impact:** Reduced processing time from 9 hours to 35 minutes with 90% less manual intervention for failed records.

5.5 Case Study 5: API Gateway and Management Layer

Objective: Create a unified entry point for 25 microservices to simplify client consumption and enhance security.

Technical Deep-Dive:

- **Technology Stack:** Ocelot, .NET Core, Azure API Management (APIM).
- **Solution:** Deployed an Ocelot gateway for internal routing and APIM for external portals, implementing gateway-level caching.
- **Impact:** Improved API performance by 20% for repeat requests and reduced partner integration time by 40%.

EDUCATION

Georgia Institute of Technology
Bachelor of Science in Computer Science

Atlanta, GA

Concentrations in Software Engineering and Database Systems.

GPA: 3.8/4.0 | Honors: Magna Cum Laude

PROFESSIONAL CERTIFICATIONS

- **Microsoft Certified: Azure Solutions Architect Expert** (AZ-305)
- **Microsoft Certified: Azure Developer Associate** (AZ-204)
- **Microsoft Certified: DevOps Engineer Expert** (AZ-400)
- **AWS Certified Developer** – Associate
- **Certified ScrumMaster (CSM)** – Scrum Alliance
- **HashiCorp Certified: Terraform Associate**

PROFESSIONAL MEMBERSHIPS & SPEAKING

- **Member, .NET Foundation:** Active contributor to open-source C# libraries.
- **Speaker, Charlotte .NET User Group:** Presented on .NET 8 Modernization (2023).
- **Speaker, Microsoft Build:** Session on Azure Container Apps Scaling (2022).
- **Contributor, StackOverflow:** Top 2% contributor in C# and .NET tags.

CONTINUOUS LEARNING & WORKSHOPS

- **Advanced Distributed Systems Design** – Udi Dahan Workshop.
- **Domain-Driven Design (DDD) Immersion** – Eric Evans Workshop.
- **Machine Learning for .NET Developers** – ML.NET Path.

VOLUNTEER EXPERIENCE

Code for Charlotte

2018 – Present

Volunteer Technical Lead

- Lead developer for a local food bank tracking application.
- Mentored 15+ career-changing developers into software roles.

PUBLICATIONS & TECHNICAL WRITING

- **Article:** "Optimizing EF Core 8 for High-Performance Systems" – 20,000+ views on Dev.to.
- **Whitepaper:** "The True Cost of Technical Debt in FinTech" – Corporate Publication (2022).
- **Open Source:** Lead Maintainer of *NetCore-Azure-Boilerplate* (1.2k+ Stars).

ADDITIONAL SOFT SKILLS & LEADERSHIP

- **Strategic Planning:** Alignment of tech roadmaps with 3-year growth objectives.
- **Stakeholder Communication:** Technical concepts translated for C-suite executives.
- **Remote Leadership:** Expert in distributed team management via structured Agile.
- **Conflict Resolution:** Consensus-based technical decision making.
- **Crisis Management:** Tiger Team lead for critical production outages.

TECHNICAL TOOLS INVENTORY

IDEs & Editors

- Visual Studio 2022 Enterprise, JetBrains Rider, VS Code, LINQPad.

Utility & Management

- Postman, SSMS, Azure Data Studio, MongoDB Compass, Docker Desktop, GitKraken.